Tiffany Taketa
BIOL 448A Directed Studies in Biology
Dr. Kaitlyn Gaynor
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Habitat Use by African Ungulates

How do spatial patterns of predation risk shapes fine-scale habitat use by seven ungulates in an African savanna?
How do specific ungulate species use tree cover?
How will their use of tree cover change in response to predation risk?

Background

- **Gorongosa National Park** is an ~4,000 km2 park in Sofala Province, Mozambique, spanning across woodland, savanna or grassland, and floodplain habitat.
- **After the Mozambiquan Civil war**, many of Gorongosa's **top predators were extirpated from the park**, except for some lions (Gaynor et al. 2016, 2021).
- In 2018, **African wild dogs** (*Lycaon pictus*), **leopards** (*Panthera pardus*), and **spotted hyenas** (*Crocuta crocuta*) were **reintroduced to the park** (Bouley et al. 2021).
- More apex predators now face the threat of extinction, globally, and **predator reintroduction is increasingly being considered** for ecological restoration (Wolf & Ripple, 2018). **Therefore, it is important to explore the regulating effects that predators have on ecosystems**.

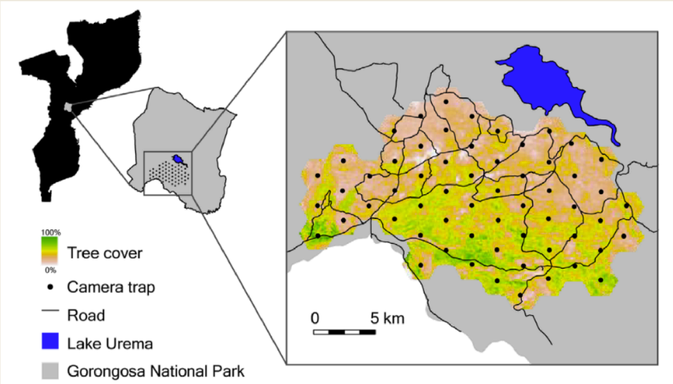

Figure 1. Map of study site showing camera trap locations represented by dots and the percent tree cover in areas across the sites as well as their location in relation to roads and Lake Urema, and an inset showing the location of the camera trap grid within the park and the park's location in Mozambique (Gaynor et al., 2021).

Table 1. Ungulate species examined in this study (Estes & Wilson, 2012; Lundgren et al., 2021).

Common name	Scientific name	Body mass (kg)	Anti-predator behavior	Feeding guild
Waterbuck	*Kobus ellipsiprymnus*	210.00	Run	Grazer
Impala	*Aepyceros melampus*	52.50	Run	Mixed-feeder
Greater kudu	*Tragelaphus strepsiceros*	213.50	Hide/Freeze	Browser
Nyala	*Tragelaphus angasii*	86.60	Hide/Freeze	Mixed-feeder
Bushbuck	*Tragelaphus scriptus*	43.25	Hide/Freeze	Browser
Oribi	*Ourebia ourebi*	17.25	Hide/Freeze	Grazer
Warthog	*Phacochoerus africanus*	82.50	Run	Grazer

Methodology

- **Camera trap survey** south of Lake Urema, in Gorongosa National Park, from 2016-2018.
- Habitat use was measured by the estimated influence of tree cover on each species' **relative activity index** (detections per trap night).
- **7 ungulate species** that are commonly detected on camera traps and represent a range of body size and feeding guild.

Research Questions & Hypothesis

Research Questions

- How does body size affect habitat use and response to predation risk?
- How does antipredator behavior affect selection for open versus closed habitat?
- How does feeding guild affect selection for open versus closed habitat?

Hypothesis: In the relative absence of predation, with the only predator being lions, feeding guild mainly drives habitat use (and outweighs other factors, e.g., body size, anti-predator behavior), except for species that are most often killed by lions in Gorongosa.

Prediction: Species most often preyed upon by lions (warthog and waterbuck) might be detected in open areas more often than other grazers in this study since they would be more responsive to predation risk.

Results & Discussion

Main finding: Most of the species preferred to be in open habitat, even in the relative absence of predators in the park
Main prediction: Habitat use will be driven by anti-predator behavior, unless species are large enough to avoid predation.

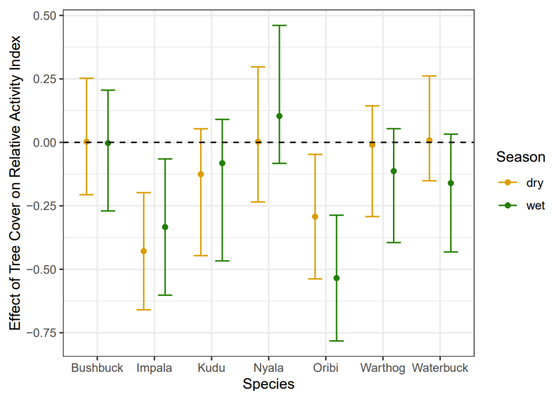

Figure 2. Estimated effect of tree cover on relative activity index from 2016-2018, during the dry season (July – September) and wet season (December – March) (Gaynor et al. 2019).

Table 2. Summary of interpretation of results and discussion, including each species' current association with tree cover and predictions for future habitat associations.

Species	Current association	Future prediction
Waterbuck	Neutral but more in open areas	Increase use of open grasslands, due to increased pressure from rising predator numbers (since they are vulnerable to introduced predators such as African wild dogs)
Impala	Open areas	Increase use of tree cover, since their main anti-predator behavior is to run or flee
Greater kudu	Neutral but more in open areas	Increase use of tree cover due to higher predation from African wild dogs, as pack sizes grow over time (African wild dogs in other savannas prefer greater kudu)
Nyala	Forested areas	Increase use of tree cover or stay relatively the same, since they are a mixed-feeder and hider but are not preferred by reintroduced predators
Bushbuck	Neutral	Increase use of tree cover, due to their anti-predator behavior (hiding in dense bush)
Oribi	Open areas	Little to no change - they are small enough to hide in tall grass, and do not need trees or shrubs
Warthog	Neutral but more in open areas	Little to no change - they are mainly predated on by lions which are already in the park

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